

# One-way TEA preservers on $c_0(\Lambda)$ and $\ell_\infty(\Lambda)$ are scalar isometries

Candidate partial-solution packet

13 August 2026

**Status:** candidate full proof of the TEA branch, likely valid; human review requested. For either  $X = c_0(\Lambda)$  or  $X = \ell_\infty(\Lambda)$ , every nonzero linear map  $T : X \rightarrow X$  which sends TEA pairs to TEA pairs is a positive scalar multiple of an isometric embedding. No boundedness or surjectivity is assumed. This completely answers the TEA version of Problem 3.7 in arXiv:2407.19276 (Problem 3.10 in the final manuscript). The parallel-only branch is not claimed.

## 1 Source question and exact scope

Let  $\mathbb{F}$  be  $\mathbb{R}$  or  $\mathbb{C}$ , and let  $\Lambda$  be an infinite index set. Vectors  $x, y$  in a normed space form a *triangle-equality-attaining pair*, abbreviated TEA, if

$$\|x + y\| = \|x\| + \|y\|.$$

Li, Tsai, Wang and Wong characterize preservers of TEA and norm-parallel pairs on finite-dimensional  $\ell_p$  spaces and obtain infinite-dimensional results under bidirectional preservation hypotheses [1]. Their Problem 3.7 asks:

Can one replace the two direction preserver conditions by that the linear map  $T$  sends parallel/TEA pairs to parallel/TEA pairs in Theorem 3.4 and Corollary 3.6, as in the finite dimensional case?

The question occurs on page 15 of the arXiv version:

*Proof.* While the implications (c)  $\implies$  (a)  $\implies$  (b) are plain, Theorem 3.5 establishes (b)  $\implies$  (c) when we identify  $\ell_\infty(\Lambda)$  with  $C(\beta\Lambda)$ . We note that  $\Lambda$  consists of all isolated points of  $\beta\Lambda$ . Thus the homeomorphism  $\sigma$  induces a bijective map  $\tau$  from  $\Lambda$  onto itself to implement (3.4). ■

*Problem 3.7.* Can one replace the two direction preserver conditions by that the linear map  $T$  sends parallel/TEA pairs to parallel/TEA pairs in Theorem 3.4 and Corollary 3.6, as in the finite dimensional case?

The authors' expanded final manuscript retains it as Problem 3.10, with the corresponding results numbered Theorem 3.7 and Corollary 3.9 [2].

This packet settles both TEA questions affirmatively and more strongly: the map need not be bounded, and in the  $\ell_\infty$  case it need not be bijective. The slash in the source also poses a distinct parallel-pair question. Its literal  $c_0$  formulation is false because every rank-one map has parallel range. The remaining rank-greater-than-one or bijective parallel question is outside the claim here. For that reason the packet is classified as a substantial partial solution to the combined source problem, although the TEA branch itself is completely solved.

## 2 Idea of the proof

First show that a nonzero one-way TEA preserver is injective. The pullback

$$q(x) := \|Tx\|_\infty$$

is then a genuine norm, possibly unrelated a priori to the original topology. The preservation hypothesis becomes a statement purely about two norms:

$$\|x + y\|_\infty = \|x\|_\infty + \|y\|_\infty \implies q(x + y) = q(x) + q(y). \quad (1)$$

Fix a point at which functions attain their sup norm, and rotate values so that the value there is nonnegative real. All functions normalized in this way form a convex cone. Equation (1) says that  $q$  is additive on each such cone, hence extends to a real-linear functional on the real span of the cone. Comparing the extensions at two points, first with phase 1 and then with phase  $-1$ , kills every possible “tail” functional. A third point forces the cone functional to be a scalar multiple of point evaluation. The scalar is independent of the point, so  $q = \gamma \|\cdot\|_\infty$ .

The decompositions are exact and algebraic. Thus no continuity assumption, finite-support approximation, or closed-graph argument is hidden in the proof.

## 3 A kernel lemma

**Lemma 1.** *Let  $X = c_0(\Lambda)$  or  $X = \ell_\infty(\Lambda)$ , where  $\Lambda$  is infinite. If a nonzero linear map  $T : X \rightarrow X$  sends TEA pairs to TEA pairs, then  $T$  is injective. No boundedness is required.*

*Proof.* Suppose  $0 \neq v \in \ker T$  and normalize  $\|v\|_\infty = 1$ . First assume that  $|v_\alpha| < 1$  for some  $\alpha \in \Lambda$ . Choose  $r > 0$  so large that  $r|v_\alpha| + 1 < r$ . The two vectors

$$rv + e_\alpha, \quad rv - e_\alpha$$

both have norm  $r$ , while their sum has norm  $2r$ ; hence they form a TEA pair. Their images are  $Te_\alpha$  and  $-Te_\alpha$ . Opposite vectors can form a TEA pair only when they vanish, so  $Te_\alpha = 0$ .

Now take any  $x \in X$  with  $x_\alpha = 0$  and choose  $r > \|x\|_\infty$ . The vectors  $re_\alpha + x$  and  $re_\alpha - x$  form a TEA pair. Their images are  $Tx$  and  $-Tx$ , so  $Tx = 0$ . Decomposing an arbitrary vector into its  $\alpha$ -coordinate and a vector with zero  $\alpha$ -coordinate gives  $T = 0$ , a contradiction.

For  $c_0(\Lambda)$  the first case always occurs. It remains to treat an  $\ell_\infty(\Lambda)$  kernel vector with  $|v_\alpha| = 1$  for every  $\alpha$ . After precomposing with a diagonal surjective isometry, we may assume that  $v = \mathbf{1}$ , so  $T\mathbf{1} = 0$ . For distinct  $\alpha, \beta$ , the vectors

$$\mathbf{1} - e_\beta, \quad e_\alpha + e_\beta$$

form a TEA pair. Applying  $T$  gives

$$\|Te_\alpha\|_\infty = \|Te_\beta\|_\infty + \|Te_\alpha + Te_\beta\|_\infty. \quad (2)$$

Interchanging  $\alpha$  and  $\beta$  yields the same identity with the first two norms exchanged. Hence  $Te_\alpha + Te_\beta = 0$ . Using three distinct indices shows  $Te_\alpha = 0$  for every  $\alpha$ . The preceding  $re_\alpha \pm x$  argument, now for arbitrary  $x$  with  $x_\alpha = 0$ , again gives  $T = 0$ . This contradiction proves injectivity.  $\square$

## 4 Rigidity on $c_0(\Lambda)$

All linear functionals in the next proof are real-linear. This is essential over  $\mathbb{C}$ , where the norm-attaining cone spans a real hyperplane.

**Lemma 2** ( $c_0$  cone rigidity). *Let  $q$  be any norm on the vector space  $c_0(\Lambda)$ , with no topological relation to  $\|\cdot\|_\infty$  assumed. Suppose (1) holds for all  $x, y \in c_0(\Lambda)$ . Then there is  $\gamma > 0$  such that*

$$q(x) = \gamma\|x\|_\infty \quad (x \in c_0(\Lambda)).$$

*Proof.* For  $i \in \Lambda$ , set

$$C_i = \{x : x_i = \|x\|_\infty \geq 0\}, \quad H_i = \{x : x_i \in \mathbb{R}\}.$$

If  $u, v \in C_i$ , then  $u, v$  form a TEA pair and  $u + v \in C_i$ . Consequently  $q|_{C_i}$  is additive and positively homogeneous. Furthermore  $C_i - C_i = H_i$ . Indeed, if  $d_i = 0$  and  $R > \|d\|_\infty$ , then

$$d = Re_i - (Re_i - d),$$

and both terms on the right belong to  $C_i$ ; real multiples of  $e_i$  are also differences of elements of  $C_i$ . Hence  $q|_{C_i}$  extends uniquely to a real-linear functional  $\Phi_i$  on  $H_i$ .

Fix  $j \neq i$  and let  $d_i = d_j = 0$ . For any unimodular scalar  $\zeta \in \mathbb{T}$  (only  $\zeta = \pm 1$  is needed), choose  $R > \|d\|_\infty$  and put

$$a = R(e_i + \zeta e_j), \quad b = a - d.$$

Both  $a, b$  lie in  $C_i$ , while  $\bar{\zeta}a, \bar{\zeta}b$  lie in  $C_j$ . Since  $d = a - b$  and norms are invariant under unimodular scalars,

$$\Phi_i(d) = \Phi_j(\bar{\zeta}d). \quad (3)$$

Taking  $\zeta = 1$  and  $\zeta = -1$  gives  $\Phi_i(d) = \Phi_j(d) = -\Phi_j(d)$ , and therefore

$$\Phi_i(d) = 0 \quad \text{whenever } d_i = d_j = 0. \quad (4)$$

Choose distinct  $j, k \neq i$ . If  $d_i = 0$ , write

$$d = (d - d_j e_j) + d_j e_j.$$

The first summand vanishes at  $i, j$ , and the second at  $i, k$ . Equation (4) shows  $\Phi_i(d) = 0$ . Thus

$$\Phi_i(x) = \gamma_i x_i \quad (x \in H_i), \quad \gamma_i = q(e_i) > 0.$$

Since  $e_i + e_j \in C_i \cap C_j$ , evaluating  $q(e_i + e_j)$  in the two cones gives  $\gamma_i = \gamma_j$ . Write the common value as  $\gamma$ .

Every nonzero  $x \in c_0(\Lambda)$  attains its norm at some coordinate  $i$ . Choose  $\zeta \in \mathbb{T}$  such that  $(\zeta x)_i = \|x\|_\infty$ . Then  $\zeta x \in C_i$ , and

$$q(x) = q(\zeta x) = \Phi_i(\zeta x) = \gamma\|x\|_\infty.$$

□

## 5 Rigidity on $C(K)$ and $\ell_\infty(\Lambda)$

**Lemma 3** ( $C(K)$  cone rigidity). *Let  $K$  be a compact Hausdorff space with at least three points, and let  $q$  be any norm on the vector space  $C(K, \mathbb{F})$ . If (1) holds for the sup norm, then  $q = \gamma\|\cdot\|_\infty$  for some  $\gamma > 0$ .*

*Proof.* For  $t \in K$ , define

$$C_t = \{f : f(t) = \|f\|_\infty \geq 0\}, \quad H_t = \{f : f(t) \in \mathbb{R}\}.$$

As before,  $q$  is additive and positively homogeneous on  $C_t$ . Also  $C_t - C_t = H_t$ . To see the only nontrivial part, let  $d(t) = 0$  and choose  $R > \|d\|_\infty$ . With  $\mathbf{1}$  the constant-one function, put

$$v = (R - |d|)\mathbf{1}, \quad u = v - d.$$

Both  $u$  and  $v$  have norm  $R$  and value  $R$  at  $t$ , and  $d = v - u$ . Together with differences of nonnegative constants, this proves the span claim. Thus  $q|_{C_t}$  extends uniquely to a real-linear functional  $\Phi_t$  on  $H_t$ .

Let  $s \neq t$  and suppose  $d(t) = d(s) = 0$ . For  $\zeta \in \mathbb{T}$ , Urysohn's lemma gives a continuous  $h_\zeta : K \rightarrow \{z : |z| \leq 1\}$  such that  $h_\zeta(t) = 1$  and  $h_\zeta(s) = \zeta$ : if  $\rho : K \rightarrow [0, 1]$  has  $\rho(t) = 0, \rho(s) = 1$ , take  $h_\zeta = (1 - \rho) + \rho\zeta$ . For  $R > \|d\|_\infty$ , set

$$v = (R - |d|)h_\zeta, \quad u = v - d.$$

Both functions have norm  $R$ ; they lie in  $C_t$ , and multiplication by  $\bar{\zeta}$  puts both in  $C_s$ . Since  $d = v - u$ , the comparison used in (3) gives

$$\Phi_t(d) = \Phi_s(\bar{\zeta}d).$$

Again  $\zeta = 1, -1$  imply

$$\Phi_t(d) = 0 \quad (d(t) = d(s) = 0). \quad (5)$$

Choose distinct  $s, r \neq t$ . If  $d(t) = 0$ , Urysohn's lemma supplies  $\rho : K \rightarrow [0, 1]$  with  $\rho(s) = 1$  and  $\rho(r) = 0$ . Then

$$d = (1 - \rho)d + \rho d.$$

The first summand vanishes at  $t, s$ , and the second at  $t, r$ . Equation (5) therefore gives  $\Phi_t(d) = 0$ . Consequently

$$\Phi_t(f) = f(t)\Phi_t(\mathbf{1}) = \gamma f(t) \quad (f \in H_t), \quad \gamma := q(\mathbf{1}) > 0.$$

The scalar is independent of  $t$ . Finally, every  $f \in C(K)$  attains its sup norm. Rotate  $f$  by a unimodular scalar so that a norming value at some  $t$  is positive real. The rotated function lies in  $C_t$ , whence  $q(f) = \gamma\|f\|_\infty$ .  $\square$

## 6 Main theorem

**Theorem 4.** *Let  $\Lambda$  be an infinite set, let  $X = c_0(\Lambda)$  or  $X = \ell_\infty(\Lambda)$  over  $\mathbb{R}$  or  $\mathbb{C}$ , and let  $T : X \rightarrow X$  be a nonzero linear map. Assume only that*

$$x, y \text{ TEA in } X \implies Tx, Ty \text{ TEA in } X.$$

*Then there exists  $\gamma > 0$  such that*

$$\|Tx\|_\infty = \gamma\|x\|_\infty \quad (x \in X).$$

*Thus  $T/\gamma$  is an isometric embedding and  $T$  is automatically bounded. If  $T$  is bijective, then  $T/\gamma$  is a surjective linear isometry.*

*Proof.* Lemma 1 makes  $q(x) = \|Tx\|_\infty$  a norm. The hypothesis gives (1). Lemma 2 proves the assertion when  $X = c_0(\Lambda)$ . For  $X = \ell_\infty(\Lambda)$ , use the standard isometric identification

$$\ell_\infty(\Lambda) = C(\beta\Lambda),$$

where  $\beta\Lambda$  is the Stone-Ćech compactification of the discrete space  $\Lambda$  [3]. Since  $\Lambda$  is infinite,  $\beta\Lambda$  has at least three points, and Lemma 3 applies. The final assertions are immediate.  $\square$

## 7 Parallel pairs, checks, and limitations

The theorem is exactly the desired one-way replacement for the TEA clauses of both source results. It is stronger than the  $c_0$  result because it does not assume reflection of TEA pairs, and stronger than the  $\ell_\infty$  result because it assumes neither reflection nor bijectivity.

For clarity, it does not prove the analogous assertion from preservation of *parallel* pairs alone. Indeed, if  $0 \neq \varphi \in c_0(\Lambda)'$  and  $0 \neq v \in c_0(\Lambda)$ , then

$$Tx = \varphi(x)v$$

has rank one and sends every pair to a parallel pair, yet it is not an isometric embedding. This disposes of the literal one-way parallel version of the  $c_0$  theorem. The source's bijective  $\ell_\infty$  parallel question, or a rank-greater-than-one reformulation, remains open in this packet.

The proof has been checked against the following edge cases.

- Zero vectors cause no problem; the nonzero hypothesis is used only to turn the kernel lemma into injectivity and to ensure  $\gamma > 0$ .
- No sequence or net is used, so uncountable  $\Lambda$  is covered.
- The function  $|d|$  is continuous in both scalar fields.
- The disk-valued Urysohn interpolation is explicit and works at non-isolated points of  $\beta\Lambda$ .
- The use of phases 1 and  $-1$  is valid over both  $\mathbb{R}$  and  $\mathbb{C}$  and is precisely what eliminates non-evaluation terms.
- No extension from finitely supported vectors, and hence no unproved continuity of  $T$ , occurs.

Human review should focus on the well-defined real-linear extension of  $q|_{C_t}$  and on the two exact cone decompositions in Lemmas 2 and 3. These are the structural heart of the argument. The kernel lemma and final reduction are elementary.

## 8 Bounded novelty search

The search through 13 August 2026 covered the run's result and attempt indexes; the arXiv id and exact title; the exact sentence "Can one replace the two direction preserver conditions"; combinations of "one-way", "TEA preserver",  $c_0$ , and  $\ell_\infty$ ; the authors' publication pages; the final 2026 JMAA manuscript [2]; and the closely related 2025–26 papers on preservation of parallel and TEA pairs. The final manuscript still labels this question Problem 3.10. No searched source states the cone rigidity lemma or Theorem 4.

Accordingly this is classified as a candidate new partial solution to the combined parallel/TEA problem, and a candidate new full solution of its TEA branch. The search is bounded rather than exhaustive; novelty confidence is moderate and specialist bibliographic review remains necessary.

## References

- [1] C.-K. Li, M.-C. Tsai, Y.-S. Wang and N.-C. Wong, *Linear maps preserving  $\ell_p$ -norm parallel vectors*, arXiv:2407.19276v1 (2024). [arXiv:2407.19276](https://arxiv.org/abs/2407.19276).
- [2] C.-K. Li, M.-C. Tsai, Y.-S. Wang and N.-C. Wong, *Linear preservers of parallel/TEA vectors in  $L_p(\mu)$  spaces*, J. Math. Anal. Appl. 553 (2026), no. 2, Article 129889. [doi:10.1016/j.jmaa.2025.129889](https://doi.org/10.1016/j.jmaa.2025.129889).
- [3] L. Gillman and M. Jerison, *Rings of Continuous Functions*, Graduate Texts in Mathematics 43, Springer, 1976.